

Material Recycling Sorter

A low-cost automated prototype that identifies and sorts three common recyclable materials into a bin; metal, glass, and paper. Accomplished using a gravity-fed chute, an array of proximity sensors, and servo-actuated gates controlled by an ESP32 microcontroller. Designed to reduce contamination at the source, improve recycling accuracy, and demonstrate sustainable engineering principles in a clear, accessible way.



SENSORS

- Inductive Proximity (IPS)**
Detects metal via eddy current damping
- Capacitive Proximity (CAP)**
Separates glass (εr 7.5) from paper (εr 2.4)
- Ultrasonic HC-SR04**
Presence detection <10 cm to trigger sequence

SDG 12 ALIGNMENT

Target 12.5 - Substantially reduce waste generation through recycling and reuse.

Target 12.8 - Ensure public awareness for sustainable development. The manual mode lets users operate gates directly, making the sorting logic tangible and educational.

AUTO MODE

6-step non-blocking state machine on ESP32

MANUAL MODE

Buttons & potentiometer for interactive demos

KEY FEATURES



Servo-Actuated Gates

4 × SG90 servos powered at 6V for increased torque, directing items into the correct bin via sequential gate opening.



3D-Printed Chute

Sorting: Metal · Glass · Paper

Gravity-fed CAD-designed structure with integrated servo housings, sensor ports, and custom horn attachments.

